Grade 11 – Term 1
C3 Practice Activity 3 with Solutions

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Evaluated criterion

C3 Applies circular-arc probability density functions to calculate interval probabilities using the geometry of circles and circular segments.

1 Introduction

Problems:

1.1. Angles on a circle: degrees and radians

1.2. Periodic variables and the cycle cut

1.3. Uniform and triangular pdf shapes

1.4. Nine-step geometric method

Criterion C2 constructed continuous pdfs and found probabilities from areas. Its uniform pdfs used rectangles, while its triangular pdfs used triangles and trapezoids. Criterion C3 applies exactly those geometric methods to variables that repeat. No new area method is needed; the new idea is to respect the cycle cut when an event passes from the end of one displayed cycle to the beginning of the next.

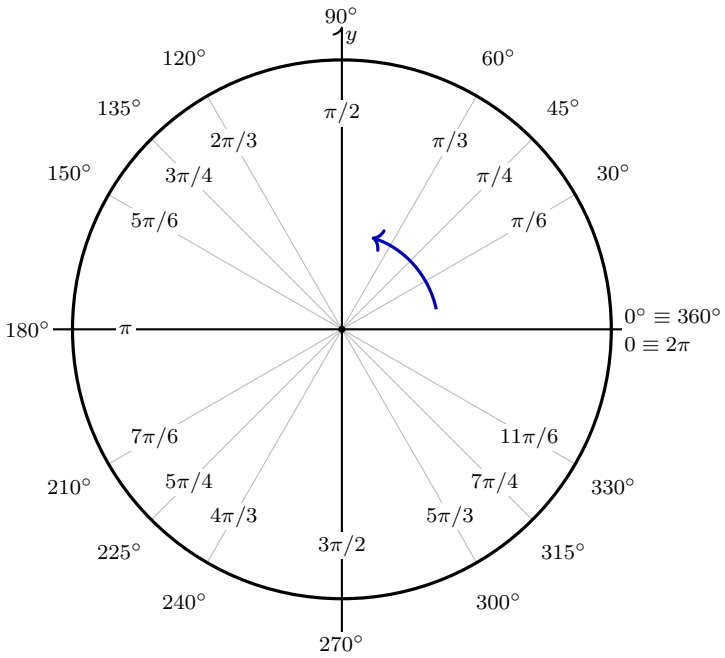
The word *circular* describes the **variable**: its values repeat after one period. The words *uniform* and *triangular* describe the **shape of its pdf**. A circular-uniform model therefore has a periodic variable and a rectangular pdf on one displayed cycle. A circular-triangular model has a periodic variable and straight sides that form one physical triangular peak around the cycle.

1.1 Angles on a circle: degrees and radians

Degrees and radians describe the same angular position using different units. One complete revolution is $360° = 2\pi$ radians, a half revolution is $180° = \pi$, and a quarter revolution is $90° = \pi/2$. The conversion rules are

$$\theta_{\text{rad}} = \theta_{\text{deg}} \frac{\pi}{180}, \qquad \theta_{\text{deg}} = \theta_{\text{rad}} \frac{180}{\pi}.$$

The standard mathematical convention measures angles counterclockwise from the positive horizontal axis, as shown below. The curved arrow marks the positive direction of rotation. Degree labels are on the outer ring and radian labels are on the inner ring so that both units remain readable.



$$0° \equiv 360°, \qquad 0 \equiv 2\pi.$$

Later compass contexts may measure angles clockwise from north rather than counterclockwise from the positive horizontal axis. This changes the chosen reference direction and direction of increase, but the degree–radian conversions remain unchanged.

The standard angles in the diagram are collected in this compact reference table.

Degrees	Radians	Degrees	Radians	Degrees	Radians
0°	0	120°	$2\pi/3$	240°	$4\pi/3$
30°	$\pi/6$	135°	$3\pi/4$	270°	$3\pi/2$
45°	$\pi/4$	150°	$5\pi/6$	300°	$5\pi/3$
60°	$\pi/3$	180°	π	315°	$7\pi/4$
90°	$\pi/2$	210°	$7\pi/6$	330°	$11\pi/6$
		225°	$5\pi/4$	360°	2π

For circular probability, angular widths can be measured in either degrees or radians, but all angles in one calculation must use consistent units. One complete cycle has width 360° or 2π, and its equivalent endpoints represent the same physical direction.

1.2 Periodic variables and the cycle cut

Compass direction repeats after one revolution, time of day repeats after one day, and rotation repeats when a device returns to its starting orientation. For example,

$$0^\circ \equiv 360^\circ, \qquad 0 \text{ h} \equiv 24 \text{ h}, \qquad f(x+P) = f(x),$$

where P is the period length. For radians, $P = 2\pi$; for degrees, $P = 360^\circ$; and for time of day in hours, $P = 24$ h.

Displaying one circle as an interval requires a *cut*. For example, $[0^\circ, 360^\circ)$ cuts the compass at north. The interval has a written beginning and end, but the physical compass has no endpoint. Thus 359° and 1° are only 2° apart across the cut. Likewise, 23:59 and 00:01 are close times even though their recorded numbers lie on opposite sides of the midnight cut.

The displayed interval is a recording convention, not a statement that the model stops beyond its endpoints. Consequently, do not automatically add an "otherwise" branch with zero density. Write the pdf on one chosen cycle and state its periodic extension. The equivalent endpoints describe the same physical value, and a wrapped event must be split only for area bookkeeping.



1.3 Uniform and triangular pdf shapes

The only area formulas required in this activity are

$$A_{\text{rectangle}} = bh, \qquad A_{\text{triangle}} = \frac{1}{2}bh, \qquad A_{\text{trapezoid}} = \frac{1}{2}(h_1 + h_2)b.$$

Here b is horizontal width and each h is a pdf height. A probability is the area of its shaded region, and the area over one complete cycle is 1.

For a uniform pdf on a period of length P, the whole cycle is a rectangle:

$$\text{Uniform: } 1 = Ph, \qquad h = \frac{1}{P}.$$

Equal-width arcs or time bands therefore have equal probabilities. This is a reasonable model when a controlled randomizer deliberately gives every part of a cycle the same chance, or when the model is explicitly introduced as a baseline for comparison. Periodicity alone does not imply uniformity.

For a triangular pdf on a period of length P, the complete physical peak has base P and height H:

$$\text{Triangular: } 1 = \frac{1}{2}PH, \qquad H = \frac{2}{P}.$$

If the mode lies inside the displayed interval, the graph rises to H and falls to zero at the cut. Straight-side heights can be found from line equations, similar triangles, or proportions, exactly as in C2.

If the physical peak lies at the cut, the unwrapped graph shows two triangular pieces: one descends from the peak at the left endpoint, and the other rises to the same peak at the equivalent right endpoint. If their bases are b_1 and b_2, then $b_1 + b_2 = P$ and

$$\underbrace{\frac{1}{2}b_1 H}_{\text{first displayed piece}} + \underbrace{\frac{1}{2}b_2 H}_{\text{second displayed piece}} = \frac{1}{2}PH = 1.$$

These are not two physical modes. They are the two displayed parts of one peak joined across the cut.



1.4 Nine-step geometric method

Use this routine for every problem.

1. Define the variable, units, period, and chosen cut.
2. Identify the rectangular or triangular shape.
3. Use total area 1 to find unknown heights.
4. Find intermediate heights with line equations, similar triangles, or proportions.
5. Write the pdf on one cycle and state its periodic extension.
6. Shade the requested region.
7. Split at the cycle cut or pdf breakpoints.
8. Add geometric areas, or subtract simple outside regions from 1.
9. Check $0 \le P(\text{event}) \le 1$ and interpret the result.

A circle has no intrinsic left or right tail. The context must name the direction and the cut. For example, "the clockwise sector from $300°$ to the end of the displayed cycle" means $[300°, 360°)$. A band within $30°$ of the cut is written as the wrapped event

$$[330°, 360°) \cup [0°, 30°].$$

The two written pieces are disjoint, so their labelled geometric areas are added.



2 Circular Uniform Distributions

Problems:

2.1. Random sensor orientation

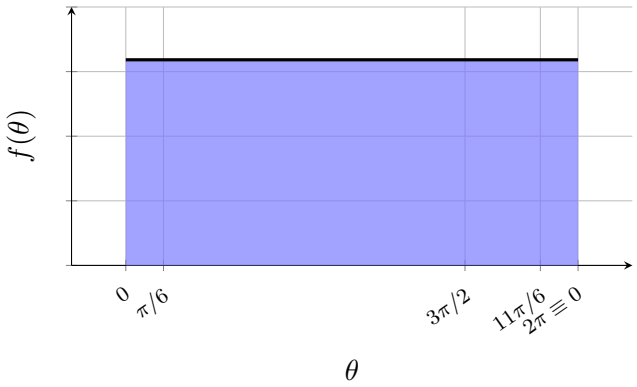
2.2. Random daily system-check time

2.3. Randomized rotating-device direction

2.1 Random sensor orientation

A university student in Bogotá is testing an augmented-reality app that places an object around the user. Let Θ be the object's clockwise orientation from the app's reference mark, measured in radians and recorded on $[0, 2\pi)$. Orientation is periodic because one full rotation returns to the same direction, so 0 and 2π are equivalent.

For each test, the app generates a direction by controlled randomization that gives equal chances to equal-width arcs. A circular-uniform pdf is therefore reasonable by design. The student checks the clockwise sector from $3\pi/2$ to the cut as one interface-test sector, and the wrapped band within $\pi/6$ of the reference mark as the app's alignment band.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of Θ from the calibration narrative, using total rectangular area 1 to find its height and stating the period.

C3.2 Calculate the interface-test tail probability for the clockwise sector from $3\pi/2$ to the end of the displayed cycle.

C3.3 Calculate the alignment-band probability within $\pi/6$ of the reference mark, crossing the cycle cut.

C3.1 - Construct the periodic pdf.

Let Θ be the app object's clockwise orientation in radians. One full turn has period 2π rad. Use $[0, 2\pi)$, cut at the app's reference mark ($0 \equiv 2\pi$).

Equal-width arcs are equally likely, so the density is a rectangle of height h. Its area over one cycle gives

$$1 = (2\pi)h \quad \Longrightarrow \quad h = \frac{1}{2\pi}\ \text{rad}^{-1}.$$

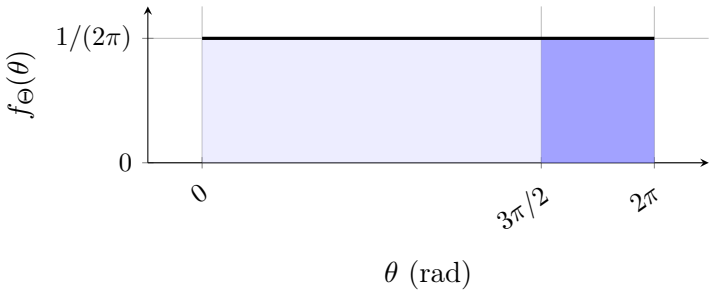
On the chosen cycle and under periodic extension, respectively,

$$f_\Theta(\theta) = \frac{1}{2\pi} \quad (0 \le \theta < 2\pi), \qquad f_\Theta(\theta + 2\pi) = f_\Theta(\theta).$$

C3.2 - Calculate the clockwise tail.

Moving clockwise from $3\pi/2$ to the reference mark gives $E = [3\pi/2, 2\pi)$ on the displayed cycle.

Interface-test sector up to the cut ($2\pi \equiv 0$)



The shaded rectangle has height $1/(2\pi)$ per radian, so its area is the favourable angular width divided by the full turn.

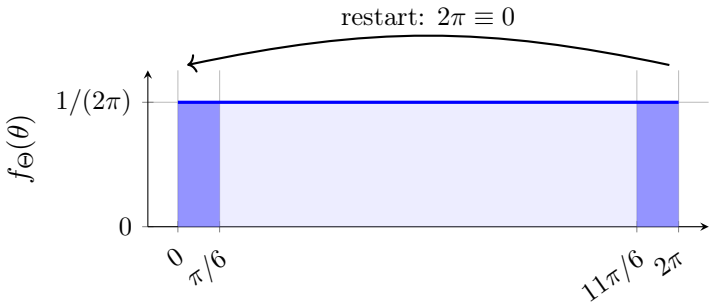
$$P(\Theta \in E) = \left(2\pi - \frac{3\pi}{2}\right)\frac{1}{2\pi} = \frac{\pi/2}{2\pi} = \boxed{\frac{1}{4}} = 25\%.$$

One quarter of the app's randomized orientations fall in the interface-test sector.

C3.3 - Calculate the wrapped alignment band.

The tolerance is $\pi/6$ rad (30°) on each side of the reference mark. In radians, the counterclockwise side is $[11\pi/6, 2\pi)$ and the clockwise side is $[0, \pi/6]$. Their union is the alignment band.

Two sides of the reference mark ($2\pi \equiv 0$)



Each shaded rectangle is $\pi/6$ rad wide and has height $1/(2\pi)$ per radian. Adding their areas gives the total favourable angular width divided by 2π.

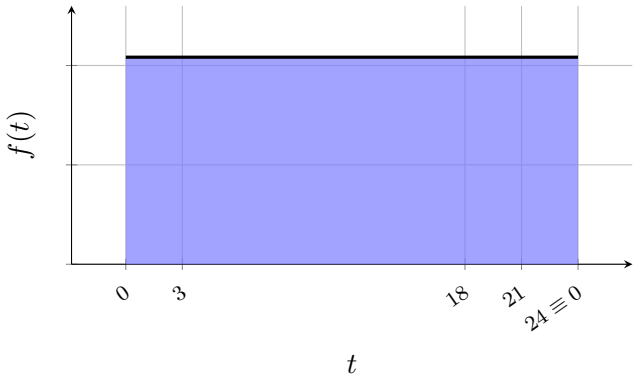
$$P(\text{alignment band}) = \frac{\pi}{6}\cdot\frac{1}{2\pi} + \frac{\pi}{6}\cdot\frac{1}{2\pi} = \frac{\pi/6 + \pi/6}{2\pi} = \boxed{\frac{1}{6}} \approx 16.7\%.$$

About 16.7% of randomized app orientations meet the alignment tolerance.

2.2 Random daily system-check time

A student team in Bogotá maintains a digital platform for a university project and schedules one automated system check each day. Let T be the check's time of day in hours, recorded on $[0, 24)$ with the cut at midnight. Time of day is periodic because the 24-hour clock returns to the same time after one day: 0 h and 24 h are equivalent.

To avoid favoring any particular time window, the scheduler uses controlled randomization that gives equal chances to equal-length intervals. This makes a circular-uniform model reasonable. The team studies the probability that a check occurs from 18:00 to midnight and the probability that it occurs from 21:00 through 03:00, when a check would fall outside typical daytime work.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of T from the scheduler narrative, using total rectangular area 1 to find its height and stating the period.

C3.2 Calculate the evening tail probability from 18:00 to the end of the displayed day.

C3.3 Calculate the overnight band probability from 21:00 through 03:00, crossing midnight.

C3.1 - Construct the periodic pdf.

Let T be the automated check's time of day in hours. The period is 24 h. Use $[0, 24)$, cut at midnight (0 h and 24 h).

Equal-length time windows are equally likely, so the density is a rectangle of height h. Its area over one day gives

$$1 = 24h \quad \Longrightarrow \quad h = \frac{1}{24}\ \mathrm{h}^{-1}.$$

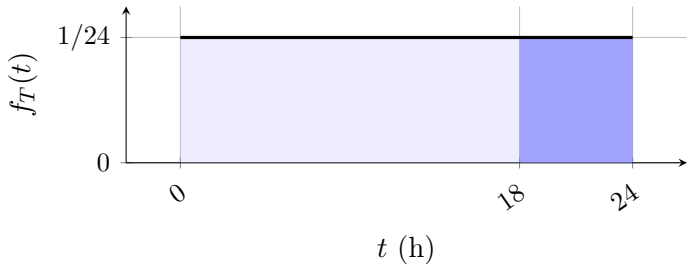
On the chosen cycle and under periodic extension, respectively,

$$f_T(t) = \frac{1}{24} \quad (0 \le t < 24), \qquad f_T(t + 24) = f_T(t).$$

C3.2 - Calculate the late-cycle tail.

In hours on $[0, 24)$, the evening tail from 18:00 to midnight is $[18, 24)$.

Evening window up to midnight $(24 \equiv 0)$



The shaded rectangle is 6 h wide and 1/24 per hour high. Its area is the favourable time length divided by 24 h.

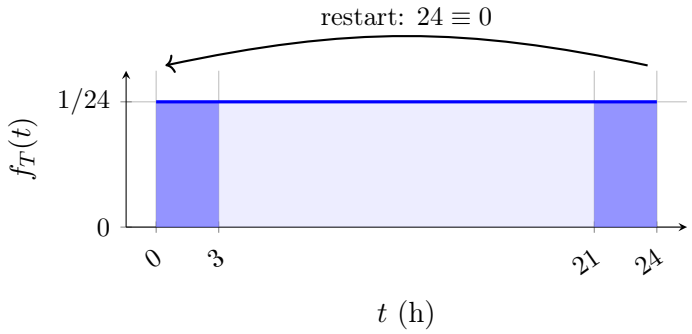
$$P(18 \le T < 24) = (24 - 18)\frac{1}{24} = \frac{6}{24} = \boxed{\frac{1}{4}} = 25\%.$$

One quarter of randomized system checks are expected between 18:00 and midnight.

C3.3 - Calculate the overnight band.

On the hour scale, 21:00 is $t = 21$ and 03:00 is $t = 3$. The overnight window runs from 21:00 to midnight and then from midnight to 03:00, giving $[21, 24) \cup [0, 3]$.

Three hours on each side of midnight $(24 \equiv 0)$



The shaded rectangles have the same height, 1/24 per hour. Their combined area is the total favourable time length divided by 24 h.

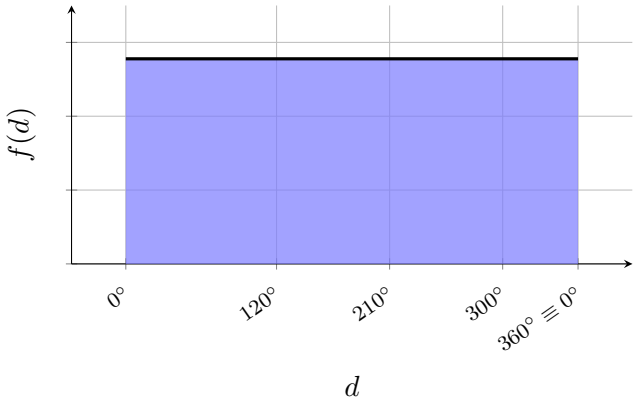
$$P(\text{overnight band}) = (24 - 21)\frac{1}{24} + (3 - 0)\frac{1}{24} = \frac{3 + 3}{24} = \boxed{\frac{1}{4}} = 25\%.$$

One quarter of randomized system checks are expected in the 21:00–03:00 window.

2.3 Randomized rotating-device direction

At a university event in Bogotá, a motorized camera is used to create 360° video clips. It records its direction D in degrees clockwise from its home mark on $[0°, 360°)$. Direction is periodic: after a 360° rotation the camera has returned to its home direction, so the displayed endpoints are equivalent.

For each clip, the controller deliberately randomizes the starting direction with no preferred sector, making a circular-uniform model appropriate by design. The clockwise sector from 300° to the cut is the side where the camera approaches its support cable, while the 120°–210° sector frames the event backdrop. The two probabilities help plan the camera setup.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of D from the controller narrative, using total rectangular area 1 to find its height and stating the period.

C3.2 Calculate the support-cable tail probability for the clockwise sector from 300° to the end of the displayed cycle.

C3.3 Calculate the backdrop-viewing band probability for the target sector from 120° through 210°.

C3.1 - Construct the periodic pdf.

Let D be the camera's starting direction in degrees clockwise from its home mark. The period is 360°. Use $[0°, 360°)$, cut at the home mark ($0° \equiv 360°$).

The controller gives equal-width sectors equal probabilities, so the density is a rectangle of height h. Its area over one turn gives

$$1 = 360h \quad \Longrightarrow \quad h = \frac{1}{360} \text{ per degree.}$$

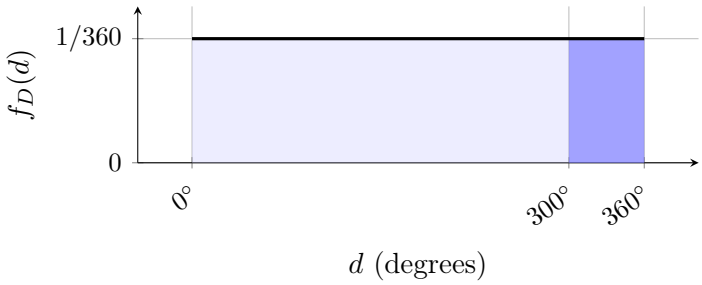
With d measured in degrees, the density on one cycle and its periodic extension are

$$f_D(d) = \frac{1}{360} \quad (0 \le d < 360), \qquad f_D(d + 360) = f_D(d).$$

C3.2 - Calculate the directional tail.

The support-cable sector runs clockwise from 300° to the home mark. On the chosen cycle it is $[300°, 360°)$.

Support-cable sector up to the cut ($360° \equiv 0°$)



The shaded rectangle has height 1/360 per degree, so its area is the favourable angular width divided by the full turn.

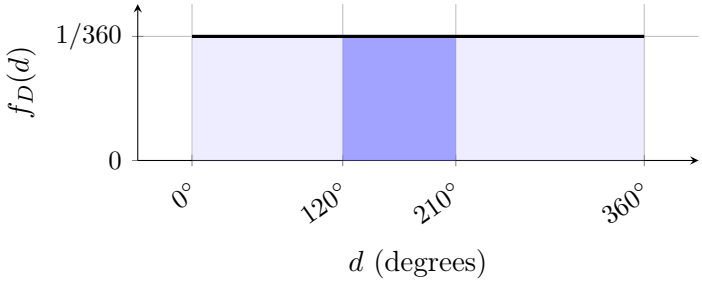
$$P(300° \le D < 360°) = (360 - 300)\frac{1}{360} = \frac{60}{360} = \boxed{\frac{1}{6}} \approx 16.7\%.$$

About 16.7% of randomized camera starts point into the support-cable sector.

C3.3 - Calculate the target sector.

The backdrop-viewing sector occupies the single interval $[120°, 210°]$.

Backdrop-viewing sector within one cycle



The shaded rectangle has height 1/360 per degree, so its area is the favourable angular width divided by the full turn.

$$P(120° \le D \le 210°) = (210 - 120)\frac{1}{360} = \frac{90}{360} = \boxed{\frac{1}{4}} = 25\%.$$

One quarter of randomized camera starts point toward the backdrop sector.



3 Circular Triangular Distributions

Problems:

3.1. Scanner direction with an interior mode

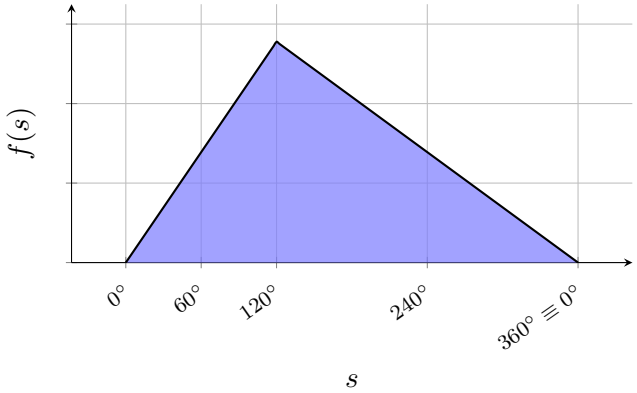
3.2. Daily peak-activity time

3.3. Direction with its peak at the cut

3.1 Scanner direction with an interior mode

In a technical-education robotics lab in Bogotá, a motorized scanner records its clockwise direction S in degrees on $[0°, 360°)$, with the cut at north. Direction is periodic because 0° and 360° are the same physical direction.

For a practice scan, the programmed profile makes north least likely, increases the direction density at a constant rate to a preferred direction of 120°, and then decreases it at a constant rate back to zero at the cut. A circular-triangular pdf is reasonable because the programmed profile has one preferred direction and straight changes on both sides. The sector from 240° to the cut faces a reflective window, while the 60°–240° band covers the project-display area being scanned.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of S from the scan profile, using total triangular area 1 to find all heights and stating the period.

C3.2 Calculate the reflective-window tail probability for the clockwise sector from 240° to the end of the displayed cycle.

C3.3 Calculate the project-display band probability from 60° through 240°.

C3.1 - Construct the periodic pdf.

Let S be the scanner's clockwise direction in degrees from north. Its period is 360°. Use $[0°, 360°)$, cut at north ($0° \equiv 360°$).

The density forms a triangle with zero height at the cut and a peak H at 120°. Its area over one turn gives

$$1 = \frac{1}{2}(360)H \quad \Longrightarrow \quad H = \frac{1}{180} \text{ per degree.}$$

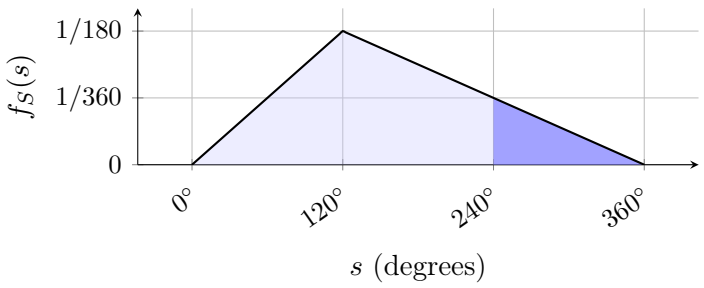
The rising side starts at zero and has slope $H/120 = 1/21600$, giving $s/21600$. The falling side ends at zero and has slope $-H/(360-120) = -1/43200$, giving $(360-s)/43200$. With s measured in degrees, the density and its extension are

$$f_S(s) = \begin{cases} \dfrac{s}{21600}, & 0 \le s \le 120, \\ \dfrac{360-s}{43200}, & 120 < s < 360, \end{cases} \qquad f_S(s+360) = f_S(s).$$

C3.2 - Calculate the reflective-window tail.

The reflective-window sector runs clockwise from 240° to north, so it is $[240°, 360°)$ on the chosen cycle.

The reflective-window tail is a triangle



The falling branch gives $f_S(240) = (360 - 240)/43200 = 1/360$. At $360° \equiv 0°$ the height is zero. The shaded triangle has area

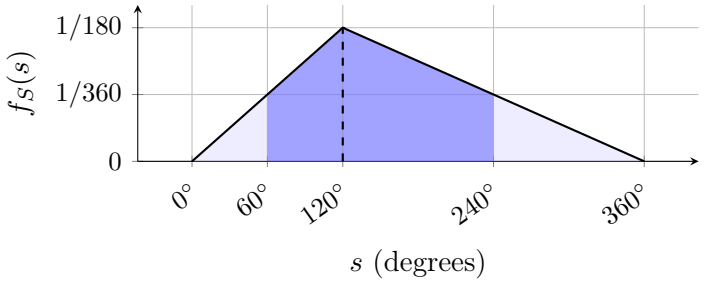
$$P(240° \le S < 360°) = \frac{1}{2}(360 - 240)\frac{1}{360} = \boxed{\frac{1}{6}} \approx 16.7\%.$$

About 16.7% of programmed scans point toward the reflective-window sector.

C3.3 - Calculate the project-display band.

The project-display band $[60°, 240°]$ crosses the peak at 120°. Split the band there into $[60°, 120°]$ and $[120°, 240°]$.

Two trapezoids meet at the 120° peak



The rising and falling branches give the boundary heights.

$$f_S(60) = \frac{60}{21600} = \frac{1}{360}, \quad f_S(120) = \frac{120}{21600} = \frac{1}{180}, \quad f_S(240) = \frac{360 - 240}{43200} = \frac{1}{360}.$$

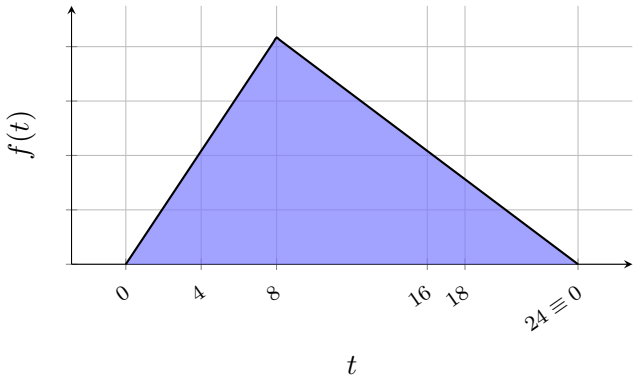
$$P(60° \le S \le 240°) = \underbrace{\frac{1}{2}\left(\frac{1}{360} + \frac{1}{180}\right)(120 - 60)}_{1/4} + \underbrace{\frac{1}{2}\left(\frac{1}{180} + \frac{1}{360}\right)(240 - 120)}_{1/2} = \boxed{\frac{3}{4}} = 75\%.$$

The scanner points toward the project-display area on three quarters of programmed scans.

3.2 Daily peak-activity time

A university study platform records T, the time of day at which its daily student activity reaches its peak, in hours on $[0, 24)$. Time of day is periodic because the clock repeats every 24 hours and midnight is both 0 h and 24 h. The model uses midnight as the cut and as the least likely peak-activity time.

For this planning model, the density rises at a constant rate from zero at midnight to a mode at 08:00, then falls at a constant rate to zero at the next midnight. A circular-triangular pdf represents this single daily peak and the specified straight changes. The platform team studies peaks after 18:00 and the 04:00–16:00 band when planning when student activity is most likely to reach its daily maximum.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of T from the activity narrative, using total triangular area 1 to find all heights and stating the period.

C3.2 Calculate the late-activity tail probability from 18:00 to the end of the displayed day.

C3.3 Calculate the daytime-activity band probability from 04:00 through 16:00.

C3.1 - Construct the periodic pdf.

Let T be the time of the platform's daily activity peak, in hours. Its period is 24 h. Use $[0, 24)$, cut at midnight (0 h and 24 h).

The density is zero at midnight and has peak height H at 08:00. The triangular area over one day gives

$$1 = \frac{1}{2}(24)H \quad \Longrightarrow \quad H = \frac{1}{12}\ \text{h}^{-1}.$$

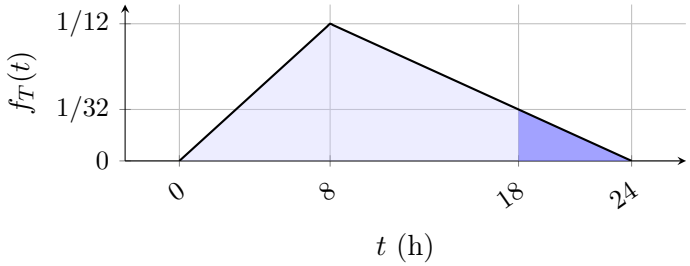
The rising side starts at zero and has slope $H/8 = 1/96$, giving $t/96$. The falling side ends at zero and has slope $-H/(24-8) = -1/192$, giving $(24-t)/192$. The density on one cycle and its extension are

$$f_T(t) = \begin{cases} \dfrac{t}{96}, & 0 \le t \le 8, \\ \dfrac{24-t}{192}, & 8 < t < 24, \end{cases} \qquad f_T(t+24) = f_T(t).$$

C3.2 - Calculate the late-activity tail.

On the hour scale, 18:00 to the midnight cut is the interval $[18, 24)$.

The late-activity tail is a triangle



The falling branch gives $f_T(18) = (24 - 18)/192 = 1/32$. At midnight, $24 \equiv 0$ and the height is zero. The shaded triangle has area

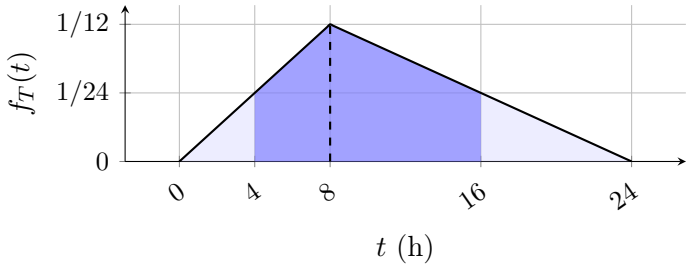
$$P(18 \le T < 24) = \frac{1}{2}(24 - 18)\frac{1}{32} = \boxed{\frac{3}{32}} = 9.375\%.$$

The platform model puts the daily peak between 18:00 and midnight on 9.375% of days.

C3.3 - Calculate the daytime-activity band.

The interval from 04:00 to 16:00 is $[4, 16]$ in hours. The density changes slope at 08:00, so split the interval into $[4, 8]$ and $[8, 16]$.

Two trapezoids meet at the 08:00 peak



The rising and falling branches give the boundary heights.

$$f_T(4) = \frac{4}{96} = \frac{1}{24}, \quad f_T(8) = \frac{8}{96} = \frac{1}{12}, \quad f_T(16) = \frac{24 - 16}{192} = \frac{1}{24}.$$

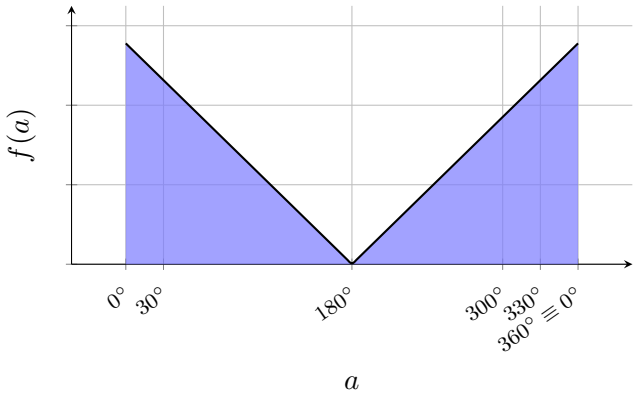
$$P(4 \le T \le 16) = \underbrace{\frac{1}{2}\left(\frac{1}{24} + \frac{1}{12}\right)(8 - 4)}_{1/4} + \underbrace{\frac{1}{2}\left(\frac{1}{12} + \frac{1}{24}\right)(16 - 8)}_{1/2} = \boxed{\frac{3}{4}} = 75\%.$$

The platform model puts the daily activity peak between 04:00 and 16:00 on three quarters of days.

3.3 Direction with its peak at the cut

At a live-music event, a programmable stage light records its direction A in degrees clockwise from north on $[0°, 360°)$. Direction is periodic, so north is both $0°$ and the equivalent endpoint $360°$; the physical direction does not stop at the recording cut.

The lighting program most strongly favors north, reduces the direction density at a constant rate to zero at south ($180°$), and then increases it at the same constant rate back toward north. A circular-triangular pdf is reasonable for this symmetric programmed preference. The sector from $300°$ to the cut covers one side of the audience area, while the band within $30°$ of north is a narrow stage-target zone.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of A, showing how two displayed triangular pieces form one physical peak at the cut, using their combined area 1 to find all heights, and stating the period.

C3.2 Calculate the audience-side tail probability for the clockwise upper-end sector from $300°$ to the end of the displayed cycle.

C3.3 Calculate the stage light's wrapped target-band probability on $[330°, 360°) \cup [0°, 30°]$.

C3.1 - Construct the periodic pdf.

Let A be the stage-light direction in degrees clockwise from north. Its period is $360°$. Use $[0°, 360°)$, cut at north ($0° \equiv 360°$).

The cut splits the physical peak H between the two displayed ends. Each triangle has base $180°$ and meets the other at zero density at south. The total area over one turn gives

$$1 = \frac{1}{2}(180)H + \frac{1}{2}(180)H = 180H \quad \Longrightarrow \quad H = \frac{1}{180} \text{ per degree.}$$

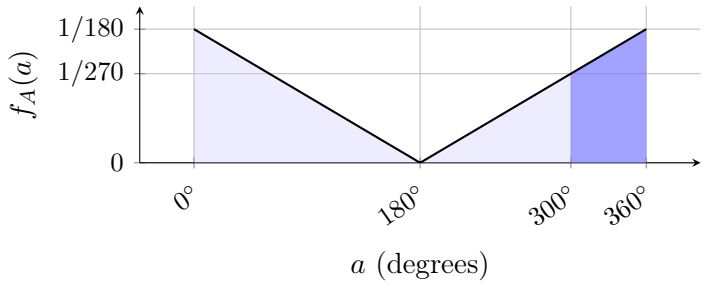
Both line segments meet at $(180, 0)$. The descending slope is $-H/180 = -1/32400$, giving $(180 - a)/32400$. The ascending slope is $H/180 = 1/32400$, giving $(a - 180)/32400$. With a measured in degrees, the density and its extension are

$$f_A(a) = \begin{cases} \dfrac{180 - a}{32400}, & 0 \le a \le 180, \\ \dfrac{a - 180}{32400}, & 180 < a < 360, \end{cases} \qquad f_A(a + 360) = f_A(a).$$

C3.2 - Calculate the upper-end directional tail.

The audience-side sector runs clockwise from 300° to the north cut. On the chosen cycle it is $[300°, 360°)$.

The audience-side trapezoid ends at north ($360° \equiv 0°$)



On the rising side, $f_A(300) = (300 - 180)/32400 = 1/270$ and the height approaches $(360 - 180)/32400 = 1/180$ at north. The shaded trapezoid has area

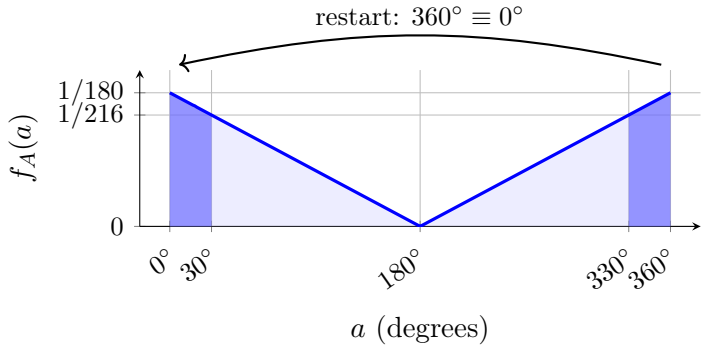
$$P(300° \le A < 360°) = \frac{1}{2}\left(\frac{1}{270} + \frac{1}{180}\right)(360 - 300) = \boxed{\frac{5}{18}} \approx 27.8\%.$$

About 27.8% of programmed stage-light directions fall in the audience-side sector.

C3.3 - Calculate the wrapped target band.

The target extends 30° ($\pi/6$ rad) to each side of north. In degrees, the counterclockwise side is $[330°, 360°)$ and the clockwise side is $[0°, 30°]$. Their union is the target band.

One trapezoid on each side of north ($360° \equiv 0°$)



At north, $f_A(0) = f_A(360^-) = 1/180$. The other boundary heights are

$$f_A(330) = \frac{330 - 180}{32400} = \frac{1}{216}, \qquad f_A(30) = \frac{180 - 30}{32400} = \frac{1}{216}.$$

Add the areas of the two 30°-wide trapezoids.

$$P(\text{target band}) = \underbrace{\frac{1}{2}\left(\frac{1}{216} + \frac{1}{180}\right)(30)}_{11/72} + \underbrace{\frac{1}{2}\left(\frac{1}{180} + \frac{1}{216}\right)(30)}_{11/72} = \boxed{\frac{11}{36}} \approx 30.6\%.$$

About 30.6% of programmed stage-light directions are within 30° of north.